

J - PIN VOLTAGE CHARTS

1993 Toyota Celica

1993 ENGINE PERFORMANCE
Toyota Pin Voltage Charts

Celica

INTRODUCTION

Pin voltage charts are supplied to reduce diagnostic time. Checking pin voltages at the ECM determines whether it's receiving and transmitting proper voltage signals. Charts may also help determine if ECM harness is shorted or opened.

NOTE: Unless stated otherwise in testing procedures, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance. Voltage readings may vary slightly due to battery condition or charging rate.

ECM LOCATION & IDENTIFICATION

ECM LOCATION TABLE

Model	Location
Celica	Bottom Center Of Dash, Under Radio

ECM PIN CONNECTOR ABBREVIATION IDENTIFICATION TABLE

Abbreviation	Function
ABS	Anti-Lock Brake ECM
ACA, ACT, ATS	A/C Amplifier
AC, AC1, A/C	A/C Compressor Or Magnetic Switch
A/D	Cruise Control Computer
AI	Air Injection Vacuum Switching Valve
AS	Air Suction Vacuum Switching Valve
BATT	Battery
BK	Brake Switch
B, +B, +B1	EFI Main Relay
CHK	Sub-Oxygen Sensor
C1	Distributor
DFG	Defogger Relay
DG	Check Connector
ECT	Electronically Controlled Transmission/Transaxle (ECT)
EGR	EGR Vacuum Switching Valve
ELS	Headlight & Defogger Relay
E1, E2, E01, E02	Engine Or Computer Ground
E11, E2, E21, E22	Sensor Ground
FC	Circuit Opening Relay
FP	Fuel Pump Relay
FPR	Fuel Pump Relay
FPU	Fuel Pressure-Up Vacuum Switching Valve
G, G1, G2	Cam Position Sensor Or Distributor
G-, G+	Cam Position Sensor Or Distributor (Crank Angle)
HT	Oxygen Sensor Heater
HT1	Oxygen Sensor Heater (Main)
HT2	Oxygen Sensor Heater (Sub)

IDL	Throttle Position Sensor
IGDA, IGDB	Ignitor
IGF	Ignitor
IG SW	Ignition Switch
IGT	Ignitor (Primary Trigger)
ISCC, ISCO	Idle Speed Control Valve
ISCV	A/C Idle-Up Vacuum Switching Valve
ISC1, ISC2, ISC3, ISC4	Idle Speed Control Valve Or Motor
KNK, KNK1, KNK2	Knock Sensor
KS	Airflow Meter
L	Neutral Start Switch Or Shift Position Switch
LP	Headlight Relay
L1, L2, L3, L4	(1) ECT
M	Neutral Start Switch Or Shift Position Switch
M-REL	EFI Main Relay
MS	Engine Oil Level Sensor
N	Neutral Start Switch Or Shift Position Switch
N/C	Clutch Switch (M/T)
NE, NE-, NE+	Cam Position Sensor Or Distributor (RPM)
NSW	Neutral/Start Switch (A/T)
No. 1-4	Injector
No. 10, 20 & 30	Injector
OD, ODT	Overdrive Solenoid
OD1	Overdrive Off Switch
OD2	Cruise Control Computer Or Overdrive Main Switch
O/G	Check Connector
OIL	Oil Pressure Switch Or A/T Oil Temperature Warning Light
OLS	Engine Oil Level Sensor
OMR	Engine Oil Feed Motor
OMT	Engine Oil Feed Motor Relay
OSFC	Overspeed Fuel Cut
OW	Engine Oil Warning Light
OX, OX1, OX+	Oxygen Sensor
OX2	Sub-Oxygen Sensor
P	Shift Position Switch Or Pattern Select Switch
PS	Power Steering ECM
PSCT	Power Steering ECM
PSW	Throttle Position Sensor
PIM	Vacuum Sensor Or Turbocharger Pressure Sensor
PSW	Throttle Position Sensor
PWR	Pattern Select Switch
R	Neutral Start Switch Or Shift Position Switch
RSC	Idle Speed Control Valve
RSO	Idle Speed Control Valve
S	Neutral Start Switch Or Shift Position Switch
SL, S1, S2	(1) ECT Solenoid
SPD, SPD1, SPD2	Vehicle Speed Sensor
SP1, SPD1	No. 1 Vehicle Speed Sensor, (1) ECT In Instrument Cluster
SP2, SPD2	No. 2 Vehicle Speed Sensor, (1) ECT In Transaxle or Transmission
STA	Starter Signal
STP	Stoplight Switch
STJ	Cold Start Injector
S1, S2, S3, S4	(1) ECT Solenoid
T, TT	Check Connector Or (2) DLC
THA, THA1	Air Temperature Sensor
THE	Evaporator Temperature Sensor
THG	EGR Gas Temperature Sensor
THO, THO1, THO2	Transmission Oil Temperature Sensor
THW	Coolant Temperature Sensor

TIL		Turbo Indicator Light
TOV		Throttle Opener Vacuum Switching Valve
TPC, TPC1		Turbocharger Pressure
		Vacuum Switching Valve
TVIS		Variable Induction System Vacuum Switching Valve
TWS		Coolant Temperature Switch
T2		Check Connector
VC		Airflow Meter, Vacuum Sensor Or
		Throttle Position Sensor
VCC		Throttle Position Sensor Or Vacuum Sensor
VF, VF1, VF2		Check Connector
V-ISC		Auxiliary Control Valve Or
		Idle-Up System Vacuum Switching Valve
VS		Airflow Meter
VSV1		Air Control Vacuum Switching Valve
VSV2		Fuel Pressure-Up Vacuum Switching Valve
VTA		Throttle Position Sensor
W		Malfunction Indicator Light
2		Neutral Start Switch Or Shift Position Switch

(1) - Electronically Controlled Transmission/Transaxle (ECT).
(2) - Data Link Connector (DLC).

PIN VOLTAGE CHARTS

ECM PIN VOLTAGE TEST

CAUTION: Perform all voltage measurements with ECM harness connector installed. Use high-impedance digital voltmeter (10,000 ohm minimum). Verify battery voltage is greater than 11 volts.

NOTE: If voltage value is not as specified, proceed to test number indicated. Tests are located in G - TESTS W/CODES article in the ENGINE PERFORMANCE section.

No.	Terminals	Condition		STD voltage (V)	TEST NO.
1	+B - E1 +B1	IG SW ON		10 - 14	②
2	BATT - E1	-		10 - 14	①
3	IDL - E2	IG SW ON	Throttle valve open	4.5 - 5.5	⑨
	VC - E2		-	4.5 - 5.5	
	VTA - E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.1 - 1.0	
			Throttle valve fully open	3.2 - 4.2	
4	VC - E2	VS - E2	-	4.5 - 5.5	⑦
			Measuring plate fully closed	3.7 - 4.3	
			Measuring plate fully open	0.2 - 0.5	
			Idling	1.6 - 4.1	
	3,000 rpm		1.0 - 2.0		
5	No.1 No.2 - E01 No.3 - E02 No.4	IG SW ON		10 - 14	⑬
6	THA - E2	IG SW ON	Intake air temp. 20°C (68°F)	1 - 3	⑥
7	THW - E2		Coolant temp. 80°C (176°F)	0.1 - 1.1	⑤
8	STA - E1	Cranking		6 - 14	⑩
9	IGT - E1	Cranking or idling		0.8 - 1.2	③
10	RSC - E1 RSO	IG SW ON	Engine ECU connectors disconnected	8 - 14	⑭
11	W - E1	No trouble ("CHECK" engine warning light off) and engine running		10 - 14	⑮
12	PIM - E2	IG SW ON		2.5 - 4.5	⑧
	VC - E2			4.5 - 5.5	
13	ACJ - E1	IG SW ON	Air conditioning ON	8 - 14	⑪

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Fig. 1: ECM Pin Voltage Test (3S-GTE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

No.	Terminals	Condition		STD voltage (V)	TEST NO.
1	+B +B1 - E1	IG SW ON		10 - 14	②
2	BATT - E1	-		10 - 14	①
3	IDL - E2	IG SW ON	Throttle valve open	10 - 14	⑨
	PSW - E2		Throttle valve fully closed	10 - 14	
4	PIM - E2	IG SW ON		3.3 - 3.9	⑦
	VCC - E2			4.5 - 5.5	
5	No.10 E01 No.20 E02			10 - 14	⑬
6	THA - E2	IG SW ON	Intake air temp. 20°C (68°F)	1 - 3	⑥
7	THW - E2		Coolant temp. 80°C (176°F)	2.0 - 2.8	⑤
8	STA - E1	Cranking		6 - 14	⑩
9	IGT - E1	Cranking or idling		0.7 - 1.0	③
10	W - E1	No trouble ("CHECK" engine warning light off) and engine running		10 - 14	⑮
11	A/C - E1	IG SW ON	Air conditioning ON	8 - 14	⑪

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Fig. 2: ECM Pin Voltage Test (4A-FE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

No.	Terminals	Condition		STD voltage (V)	TEST NO.
1	+B +B1 - E1	IG SW ON		10 - 14	②
2	BATT - E1	—		10 - 14	①
3	IDL - E2	IG SW ON	Throttle valve open	8 - 14	⑨
	VC - E2		—	4.5 - 5.5	
	VTA - E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8 - 1.2	
			Throttle valve fully open	3.2 - 4.2	
4	PIM - E2	IG SW ON		3.3 - 3.9	⑦
	VC - E2			4.5 - 5.5	
5	No.10 - E01 No.20 - E02				
6	THA - E2	IG SW ON	Intake air temp. 20°C (68°F)	1.9 - 2.9	⑥
7	THW - E2		Coolant temp. 80°C (176°F)	0.1 - 1.1	⑤
8	STA - E1	Cranking		6 - 14	⑩
9	IGT - E1	Cranking or idling		0.8 - 1.2	③
10	ISCC - E1 ISCO - E1	IG SW ON	Engine ECU connectors disconnected	8 - 14	⑭
11	W - E1	No trouble ("CHECK" engine warning light off) and engine running		10 - 14	⑮

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Fig. 3: ECM Pin Voltage Test (5S-FE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

ECM WIRING HARNESS RESISTANCE TEST

CAUTION: When measuring resistance at ECM wiring harness, DO NOT touch ECM terminals with ohmmeter. Turn ignition off and unplug ECM wiring harness connector. Tester probe should be inserted into the wiring connector from the WIRING side.

Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed	2,300 or less
VTA - E2	Throttle valve fully open	3,100 - 12,100
	Throttle valve fully closed	470 - 6,100
VC - E2	-	3,900 - 9,000
VS - E2	Measuring plate fully closed	200 - 600
	Measuring plate fully open	20 - 1,200
THA - E2	Intake air temp. 20°C (68°F)	2,000 - 3,000
THW - E2	Coolant temp. 80°C (176°F)	200 - 400
G1 G2 - G $\ominus$	Cold	125 - 190
NE - G $\ominus$	Cold	155 - 240
RSC +B RSO - +B1	-	19.3 - 22.3

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Fig. 4: ECM Wiring Harness Resistance Test (3S-GTE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed	0
PSW - E2	Throttle valve fully open	0
	Throttle valve fully closed	Infinity
THA - E2	Intake air temperature 20°C (68°F)	2,000 - 3,000
THW - E2	Coolant temperature 80°C (176°F)	200 - 400
G1 NE - G $\ominus$	Cold	185 - 265

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Fig. 5: ECM Wiring Harness Resistance Test (4A-FE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

Resistance of ECU Wiring Connectors (5S-FE)

Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed (Throttle opener must be cancelled first)	2,300 or less
VTA - E2	Throttle valve fully open	2,000 - 10,200
	Throttle valve fully closed (Throttle opener must be cancelled first)	200 - 5,700
VC - E2	—	2,500 - 5,900
THA - E2	Intake air temp. 20°C (68°F)	2,000 - 3,000
THW - E2	Coolant temp. 80°C (176°F)	200 - 400
G+ - G-	Cold	185 - 265
NE+ - NE-	Cold	370 - 530
ISCC +B ISCO - +B1	—	19.3 - 22.3

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Fig. 6: ECM Wiring Harness Resistance Test (5S-FE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

ECM TERMINAL IDENTIFICATION

E01	No. 1	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	OX	PIM	THW	THA	VS	VC	STA	AC	SPD	ATS	FPR	W	STP	ELS	BATT
E02	No. 3	No. 4			IGT		G1	G⊖			E1	TE1	TE2	KNK	IDL	VTA	THG	E2	ACT						FC	+B1	+B

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Fig. 7: ECM Terminal ID (3S-GTE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

E01	No. 10	STA	OX	G ⊖	G	IGF	IGT	THA	PIM	THW	NSW	EGR	T	ACT			FC		BATT + B1
E02	No. 20	E1		E21	NE	THG	IDL	VCC	PSW	E2	OD or HT	V- ISC	VF	ODT			SPD A/C		W + B

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Fig. 8: ECM Terminal ID (4A-FE)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

E01	No. 10	No. 20	ISCO	ISCC	ISCV	2	NE -	NE +	IGF	S1	SL	VF	TT	OX1	OX2	THW	THA	PIM	VC	STA	SPD	ACA	OD2	W	B/K	THE	ELS	BATT
E02			EGR	P	IGT	L	G +	G -	SP2	S2	E1	E21	TE1	TE2	KNK	IDL	VTA	THG	E2	NSW	ACT	OD1			ATS	FC	+B1	+B

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Fig. 9: ECM Terminal ID (5S-FE Automatic Transaxle)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

E01	No. 10	ISCO	ISCC	ISCV	G +	NE +	IGF	STA	ISCV	VF	TE2	OX1	KNK	THW	THA	PIM	OX2	ACT	FC	ACA	BATT	+B1
E02	No. 20	E1	EGR	IGT	G -	NE -				E21	TE1	THE	THG	IDL	VC	VTA	E2	ELS	SPD	ATS	W	+B

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Fig. 10: ECM Terminal (5S-FE Manual Transaxle)
Courtesy of Toyota Motor Sales, U.S.A., Inc.

K - SENSOR RANGE CHARTS

1993 Toyota Celica

1993 ENGINE PERFORMANCE
Toyota Sensor Operating Range Charts
Celica

INTRODUCTION

Sensor operating range information can help determine if a sensor is out of calibration. An out-of-calibration sensor may not set a trouble code, but it may cause driveability problems.

NOTE: Unless stated otherwise in testing procedure, perform all voltage tests using a Digital Volt-Ohmmeter (DVOM) with a minimum 10-megohm input impedance.

AIRFLOW METER

NOTE: Airflow meter may be referred to as Volume Airflow (VAF) meter or Mass Airflow (MAF) meter.

AIRFLOW METER RESISTANCE SPECIFICATIONS TABLE

Application & Terminals (1)	Ohms
Turbo	
E2-VC	200-400
E2-THA	
-4°F (20°C)	10,000-20,000
32°F (0°C)	4000-7000
68°F (20°C)	2000-3000
104°F (40°C)	900-1300
140°F (60°C)	400-700
E2-VS	
Measuring Plate Fully Closed	200-600
Measuring Plate Fully Opened	20-1200

(1) - For terminal identification and testing, see I - SYS /COMP TESTS article in the ENGINE PERFORMANCE section.

AIR TEMPERATURE SENSOR

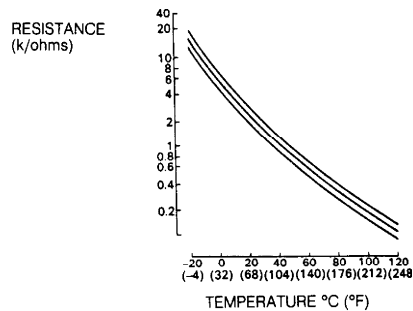
NOTE: Air temperature sensor may be referred to as intake air temperature sensor.

1.6L 4A-FE & 2.2L 5S-FE

Air temperature sensor is located in air intake, near air filter. Measure resistance between sensor terminals. For resistance specifications, see Fig. 1.

2.0L Turbo

Air temperature sensor is an integral part of airflow meter. Check resistance at terminals E2 and THA of airflow meter. See AIRFLOW METER RESISTANCE SPECIFICATIONS table.



CAMRY 2.2L, CELICA 1.6L 4A-FE & 2.2L 5S-FE,
 PASEO & TERCEL 93B79583

Fig. 1: Checking Air Temperature Sensor (1.6L)
 Courtesy of Toyota Motor Sales, U.S.A., Inc.

COLD START INJECTOR TIME SWITCH

COLD START INJECTOR TIME SWITCH RESISTANCE SPECIFICATIONS TABLE

Application & Terminals (1)	Temperature °F (°C)	Ohms
Turbo		
STA-STJ	Less Than 50 (10)	30-50
	Greater Than 77 (25)	70-90
STA-Ground		30-90

(1) - For terminal identification and testing, see I - SYS/
 COMP TESTS article in the ENGINE PERFORMANCE section.

COOLANT TEMPERATURE SENSOR

NOTE: Coolant temperature sensor may also be referred to as Engine
 Coolant Temperature (ECT) sensor.

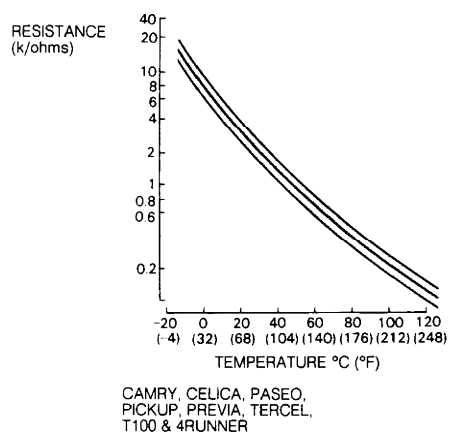
Measure coolant temperature sensor resistance between sensor
 terminals. For resistance specifications, see Fig. 2.

EGR GAS TEMPERATURE SENSOR

EGR GAS TEMPERATURE SENSOR SPECIFICATIONS TABLE (1)

Temperature - °F (°C)	k/Ohms
122 (50)	69-89
212 (100)	11-15
302 (150)	2-4

(1) - For terminal identification and testing, see I - SYS/
 COMP TESTS article in the ENGINE PERFORMANCE section.



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Fig. 2: Checking Coolant Temperature Sensor
Courtesy of Toyota Motor Sales, U.S.A., Inc.

MANIFOLD ABSOLUTE PRESSURE SENSOR

NOTE: See VACUUM SENSOR section in this article.

OXYGEN SENSOR HEATER

OXYGEN SENSOR HEATER RESISTANCE SPECIFICATIONS TABLE (1)

Application	Ohms
Celica	5.1-6.3 @ 68°F (20°C)

(1) - Measure resistance between sensor terminals +B and HT.
For terminal identification, see I - SYS/COMP TESTS
article in the ENGINE PERFORMANCE section.

THROTTLE POSITION SENSOR

NOTE: For terminal identification and testing procedures, see
I - SYS/COMP TESTS article in the ENGINE PERFORMANCE section.

TPS RESISTANCE SPECIFICATIONS TABLE

Application	Clearance In. (mm)	Terminal	Ohmmeter Reading
1.6L 4A-FE	.024 (.61)	IDL & E2	Continuity
		PSW-E2	No Continuity
		IDL-E2	No Continuity
	.031 (.79)	PSW-E2	No Continuity
		IDL-E2	No Continuity
2.0L Turbo (1) ...	Fully Open	PSW-E2	Continuity
		VTA & E2	470-6100
	0 (0)	IDL & E2	2300 Or Less
	.020 (.51)	IDL & E2	No Continuity
	.028 (.71)	VTA & E2	3100-12,100
2.2L 5S-FE (1)	Fully Open ...	VC & E2	3900-9000
		VTA & E2	200-5700
	0 (0)	IDL & E2	2300 Or Less
	.020 (.51)	IDL & E2	No Continuity
	.028 (.71)	VTA & E2	2000-10,200
	Fully Open ...	VC & E2	2500-5900

(1) - Apply vacuum to throttle opener before checking TPS.

TURBOCHARGING PRESSURE SENSOR

TURBOCHARGING PRESSURE SENSOR OUTPUT VOLTAGE SPECIFICATIONS TABLE

Applied Vacuum In. Hg	Output Voltage Drop
3.94	.15-.35
7.87	.40-.60
11.81	.65-.85
15.75	.90-1.10
19.69	1.15-1.35

VACUUM SENSOR

NOTE: Vacuum sensor may also be referred to as Manifold Absolute Pressure (MAP) sensor. Vacuum sensor is used on 1.6L 4A-FE and 2.2L 5S-FE. For testing procedures, see I - SYS/COMP TESTS article in the ENGINE PERFORMANCE section.

VACUUM SENSOR SUPPLY VOLTAGE SPECIFICATIONS TABLE

Application	Voltage
1.6L 4A-FE & 2.2L 5S-FE	4.50-5.50

VACUUM SENSOR OUTPUT VOLTAGE DROP SPECIFICATIONS TABLE

Applied Vacuum In. Hg	Output Voltage Drop
3.94	.3-.5
7.87	.7-.9
11.81	1.1-1.3
15.75	1.5-1.7
19.69	1.9-2.1

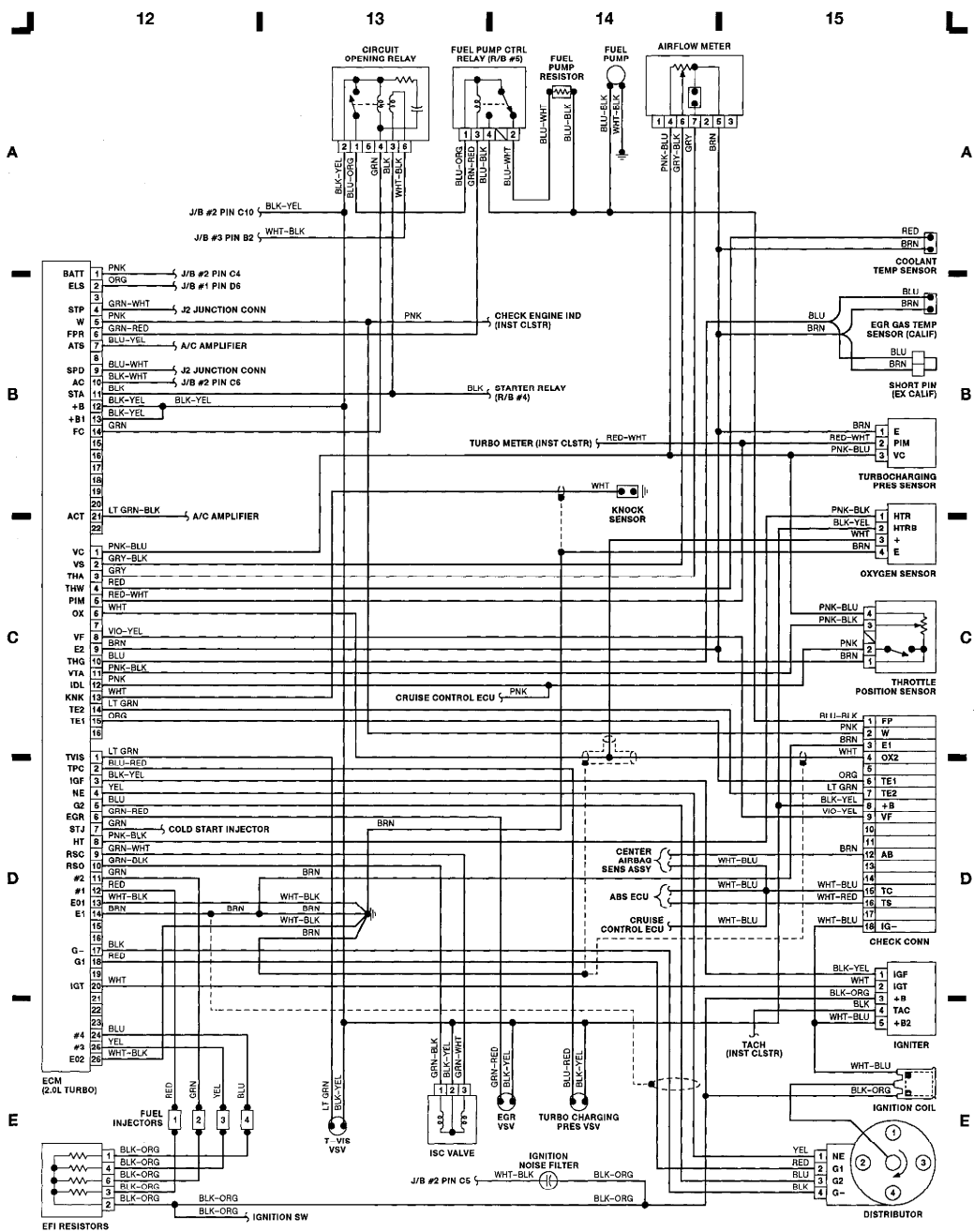
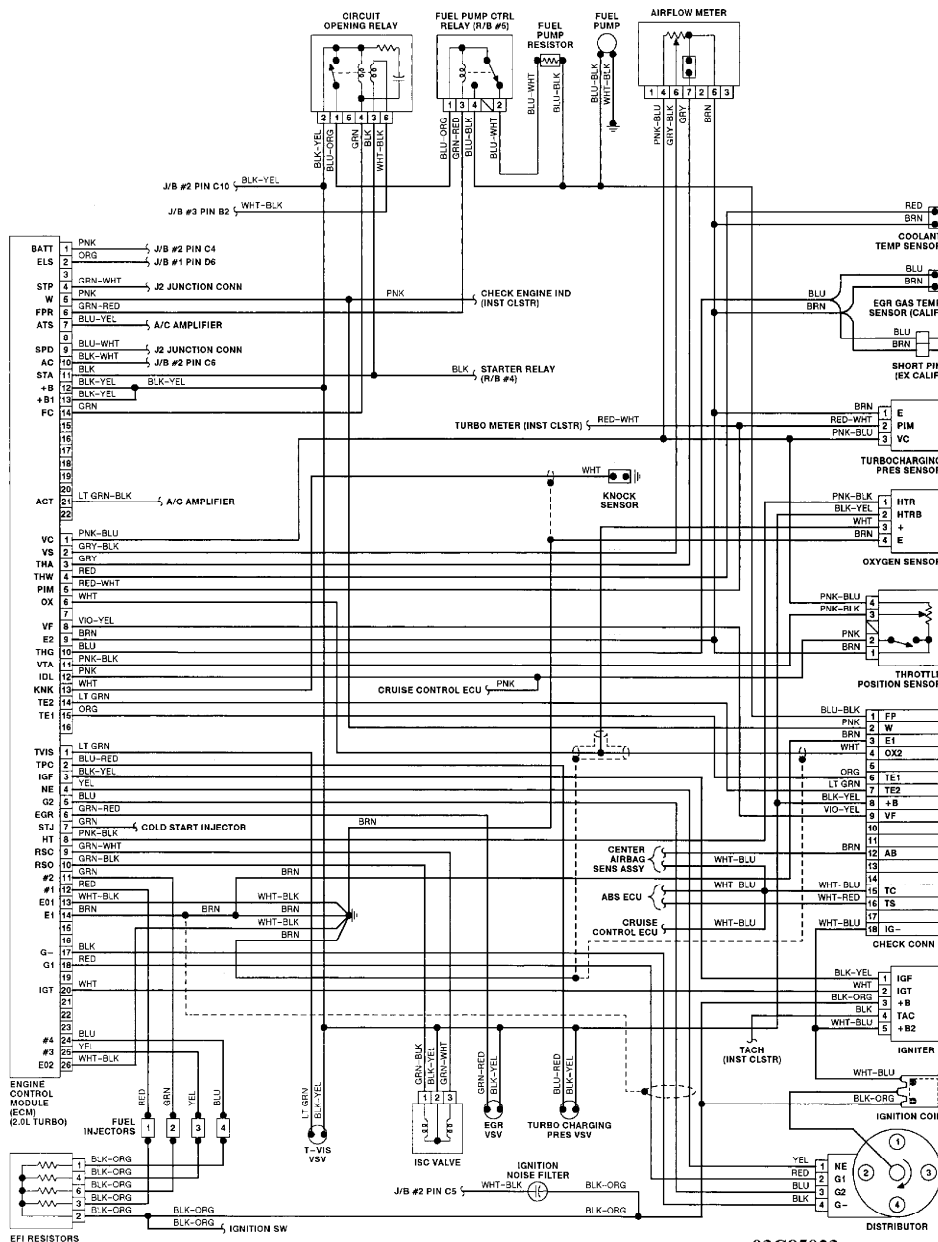


Fig. 4: ECM ECU & Engine Sensors (2.0L Turbo) (Grid 12-15)



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Fig. 2: Wiring Diagram (Celica 2.0L Turbo)